



FROM PROMPTS TO PREDICTIONS:
FORECASTING PSYCHOLOGICAL
DISORDERS WITH LARGE LANGUAGE
MODELS AND MACHINE LEARNING

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Abstract

Psychological disorders affect a significant percentage of the global population, contributing to a global healthcare burden. Large language models (LLMs) have demonstrated remarkable capabilities in many fields, yet their potential for forecasting psychological disorder onset from numerical data remains unexplored. This thesis compared the performance of three approaches to forecast psychological disorder onset between two time points, using an exemplary psychological dataset. We evaluated six machine learning (ML) algorithms, six leading LLMs with six prompting strategies, and three unification methods. We also used LLMs to explain misclassifications. Performance was assessed using accuracy, sensitivity, specificity, macro F1 score, and Matthews Correlation Coefficient (MCC). ML models demonstrated higher accuracy and specificity, with Naive Bayes performing best overall, achieving an accuracy of 0.74 and MCC of 0.24. LLMs showed superior sensitivity for disorder onset detection, but lower specificity and accuracy. The best performing LLM was GPT with profiled simple prompting, achieving an accuracy of 0.72 and MCC of 0.18. The best unification approach used LLM predictions as ML predictors and performed similarly to using the original predictor set. LLMs successfully categorized misclassification reasons, with "lack of context" most frequent. LLMs hold great potential for psychological disorder forecasting through their ability to process numerical, real-time data and superior sensitivity, but require careful implementation due to limitations in transparency and reproducibility.

Chapter 1

Introduction

Psychological disorders are a global health concern with severe consequences. Approximately 50% of the global population is likely to experience the onset of at least one psychological disorder by the age of 75, with onset occurring during late adolescence or early adulthood (McGrath et al., 2023). These psychological disorders contribute significantly to the growing burden on healthcare systems worldwide (GBD 2019 Mental Disorders Collaborators, 2022), causing difficulties in a variety of important areas of life (World Health Organization, 2022a, 2022b, 2022c). These include disturbances in an individual’s cognition, emotional regulation, and behavior, as well as psychological distress, impairment in functioning, and risk of self-harm. More generally, psychological disorders can lead to problems with relationships, employment, and education. Furthermore, psychological disorders are frequently comorbid with physical health conditions and associated with an increased risk of poor clinical outcomes in various physical diseases (Dragioti et al., 2023; Jürisson et al., 2021; Wienand et al., 2024). Individuals with lifetime psychological disorders experienced lower total positive mental health, lower overall health, and lower quality of life (Vaingankar et al., 2020).

Given this substantial burden, the development of reliable and precise clinical prediction models is essential for improving the detection, treatment, and monitoring of psychological disorders (Rosenfeld et al., 2019). While machine learning (ML) has become more popular in psychological research, researchers still face practical challenges when using ML methods on psychological data (Lavelle-Hill, Smith, & Murayama, 2025). The complex nature of psychological disorders makes forecasting particularly challenging, underlining the need for innovative approaches for psychological disorder detection.

In this context, large language models (LLMs) have demonstrated remarkable capabilities beyond their original natural language processing (NLP) skills. They excel at a wide range of tasks, including mathematics, coding, vision, medicine, law and psychology (Bubeck et al., 2023). However, to the best of our knowledge,

their potential for forecasting psychological disorder incidence from numerical data remains unexplored. This gap poses the question what capabilities and limitations LLMs have with regard to forecasting psychological disorder onsets.

1.1 Psychological Data Characteristics and Prediction Methods

Psychological data often involve specific characteristics that present analytical challenges and can constrain statistical approaches. These datasets are often characterized by limited sample sizes and missing data (Kyriazos, 2018; Lavelle-Hill et al., 2025). Additionally, psychological data often involve latent constructs that cannot be measured directly. Psychological items quantifying these constructs are often interpreted subjectively and may be affected by measurement error. Psychological data are frequently collected from the same individuals across multiple time points. This creates a longitudinal setting with clustered observations that violate independence assumptions of traditional statistical methods (Lavelle-Hill et al., 2025). Furthermore, the distribution of psychological disorders creates highly unbalanced settings: In 2019, 12% of the global population were living with a psychological disorder (World Health Organization, 2022a). In these unbalanced datasets, the majority class (healthy people) outnumber positive cases (people with a psychological disorder). These aspects of psychological data present challenges for the accuracy and generalizability of predictive models.

Current statistical methods used for data analysis in psychology reflect these difficulties. A review from 2018 shows that in psychology the most frequently used procedures were ANOVA for parametric tests and regression models for modeling relationships between variables (Blanca, Alarcón, & Bono, 2018). Psychological studies often used linear, logistic or Poisson regression to predict psychological disorders such as depression and eating disorders (Bornioli, Lewis-Smith, Slater, & Bray, 2021; Hazzard et al., 2021; Piccin et al., 2025; Waszczuk et al., 2022). However, these models often require strict assumptions, such as linearity, independence of observations, normality of residuals, or homoscedasticity.

In recent years, more advanced ML approaches for psychological prediction have increased and demonstrated promising results (Lavelle-Hill et al., 2025). While psychological research traditionally focused on explaining the causes of behavior, ML enables researchers to shift their focus towards improving the prediction of behavior (Yarkoni & Westfall, 2017). In this sense, explanatory approaches aim to compare the average risk of an outcome between different groups within a population, whereas prediction models aim to maximize the ability to distinguish

between individuals with and without a clinical outcome, such as a psychological disorder (Meehan et al., 2022). In longitudinal settings, ML enables to forecast future events by analyzing the health trajectories of patients and predicting future outcomes with high accuracy (Rajkomar, Dean, & Kohane, 2019).

These ML approaches can partially relax the strict assumptions that traditional psychological methods impose. However, they still require large datasets, which can be particularly difficult to obtain in psychological research. Despite this, ML models were able to predict several psychological outcomes, including various psychological disorders (Bracher-Smith, Crawford, & Escott-Price, 2021), recovery from psychological disorders (Bokma et al., 2022), therapy outcomes (Hornstein, Forman-Hoffman, Nazander, Ranta, & Hilbert, 2021), and treatment outcomes (Sajjadian et al., 2021), using diverse sets of predictors and ML methods.

To conclude, challenges in psychological data often include limited and unbalanced datasets, which potentially contain missing data, measurement errors and longitudinal predictors. Typical challenges in traditional psychological prediction methods include the restrictive selection of models and the strict assumptions made by these models. Given these challenges, alternative approaches for psychological prediction should be explored. The recent developments in NLP and LLMs present a new opportunity to address some of these challenges, potentially offering a new method for forecasting psychological outcomes.

1.2 The Emergence of LLMs in Healthcare and Psychology

NLP is "a field of artificial intelligence (AI), computational linguistics, and computer science" (Fanni, Febi, Aghakhanyan, & Neri, 2023). It can be used to understand or generate human texts and languages. NLP models a probability distribution over a sequence of discrete tokens, which can be words, characters or bytes. The introduction of transformers in 2017 led to the development of LLMs as a subsection of NLP. These models use self-attention to weigh the influence of different words in a sequence and compute an overall representation of it (Vaswani et al., 2017). LLMs are neural networks with billions of parameters that predict the next probable word in sequence of text (Sartori & Orrù, 2023). They are based on self-supervised learning and trained on huge datasets from the internet.

Typical tasks of NLP include processing and analyzing text, such as machine translation, language modeling, text classification, text generation, and information retrieval (Patwardhan, Marrone, & Sansone, 2023). Interactive conversational agents are one of the most recent applications, with models like ChatGPT demon-

strating remarkable capabilities across many NLP tasks (Caruccio et al., 2024). In particular, Generative Pre-trained Transformers (GPT) show capacity for solving tasks without explicit training. They show reasoning abilities similar to humans, enabling them to solve a wide range of analogy problems (Webb, Holyoak, & Lu, 2023). Furthermore, they show human-like content effects on reasoning tasks (Dasgupta et al., 2024). OpenAI’s GPT-4 model can solve tasks that related to mathematics, coding, vision, medicine, law, and psychology (Bubeck et al., 2023).

1.2.1 NLP in Clinical Research

NLP has been used for a wide range of tasks in medical and clinical studies. Researchers are increasingly using NLP and LLMs to enhance clinical research and diagnoses (Dennstädt et al., 2024; Haider et al., 2024). They can facilitate clinical decisions, increase speed, and create effective treatments (Agharia, Sza-tkowski, Fraval, Stevens, & Zhou, 2024; Caruccio et al., 2024). LLMs were used to automatically classify oncological clinical trials (Dennstädt et al., 2024) and breast conditions using medical classification systems (Haider et al., 2024), as well as to formulate orthopaedic decisions (Agharia et al., 2024). They have also been used in radiology and to interpret clinical images (Caruccio et al., 2024). LLMs could predict patient outcomes, improve hospital triage procedures, and create diagnostic models for the early detection of chronic diseases (Locke et al., 2021). These previous studies have demonstrated the potential of LLMs in clinical research. LLMs can analyze and classify patient data to make clinical decisions. For example, ChatGPT can evaluate symptoms, generate diagnoses and provide information about medical conditions (Caruccio et al., 2024).

1.2.2 NLP in Psychological Research

NLP has also emerged in the field of clinical and cognitive psychology. Research on NLP and psychology has revealed that LLMs can tackle many reasoning and problem-solving tasks from cognitive psychology without being explicit training. Models like GPT-3.5 performed at neurotypical human levels across tests measuring human intelligence and cognitive impairment (Sartori & Orrù, 2023).

LLMs displayed cognitive ability in many settings and demonstrated complex psychological reasoning. Recent studies show LLMs’ abilities in psychological text analysis. GPT-3.5 can solve vignette-based tasks as successfully as humans and can make decisions based on descriptions (Binz & Schulz, 2023). GPT performed as well as or better than dictionary analysis and fine-tuned ML models at detecting psychological constructs (i.e., sentiment, discrete emotions, offensiveness, and moral foundations) in tweets and news headlines (Rathje et al., 2024). GPT is also

used as an automated text analysis tool, showing high performance in tasks for sentiment and thinking style analysis or offensive language and emotion detection in text (Abdurahman et al., 2024). NLP and LLMs have also emerged as helpful tools in the area of clinical psychology. NLP technologies have supported psychological applications by detecting suicide risk, analyzing psychotherapy sessions, and identifying patient emotions (Stade et al., 2024).

To conclude, previous research showed that LLMs have the potential to detect psychological constructs from text, make cognitive decisions, and exhibit reasoning capabilities comparable to humans. Notably, the models were only being trained to predict the probability of the next word, suggesting that LLMs capture human cognitive abilities despite no explicit training on psychological tasks. Still, clinical psychology represents a high-stakes environment requiring expertise and careful implementation. LLMs hold promise for psychological classification and diagnosis. Nevertheless, their potential for forecasting psychological disorders from structured numerical data remains unexplored.

1.3 The Unexplored Frontier of LLMs for Psychological Prediction

Despite LLMs having the potential to be applied in clinical, diagnostic and psychological areas, current NLP applications mostly use cross-sectional text data. LLMs have demonstrated remarkable capabilities in language-dependent tasks, such as cognitive testing, classification of clinical notes, analysis of therapy transcripts, and detection of psychological constructs. However, traditional psychological disorder forecasting primarily relies on quantitative variables that lack an initial textual structure. These structured, numerical data include for example demographic variables, self-reports, clinical assessments, and longitudinal observations measured over time. The data are typically stored in tabular formats. There is further research needed to investigate the potential of LLMs to forecast psychological disorder onsets using numerical data. This gap at the intersection of AI and clinical psychology can be divided into four areas.

Psychological Application Gap The use of LLMs as prediction models has been investigated in various fields, including medical research. LLMs have demonstrated abilities of cognitive reasoning, decision making, and clinical diagnoses. However, their application in clinical psychology and in forecasting psychological disorder onset remains unexplored. This raises the question of whether LLMs can be used as prediction models in the field of psychology.

Implementation Gap Previous psychological research has mostly relied on statistical methods and ML algorithms to study the relationship between predictors and targets. However, it remains unclear whether LLMs can efficiently process structured, numerical psychological data when it is converted into textual prompts. There has been little research into which LLM provider and which prompting strategy work best for this task. This raises the question of which LLM implementation is most effective in forecasting psychological disorder onset.

Comparison and Performance Gap There has been no systematic comparison between the performances of ML algorithms and LLM-based methods for predicting psychological disorder onset. There are currently only few performance standards that quantify the abilities of LLMs in the area of psychological prediction. This raises the question of whether LLMs can compete with ML models on this task, and how effective unification approaches of ML and LLMs are.

Explainability Gap Although this thesis focuses on precise prediction rather than explaining the causes of outcomes, interpretable models are crucial for psychological decision-making. The "black box" nature of LLMs is particularly challenging in terms of interpretability and explainability. While traditional ML methods can use feature importance to improve explainability, LLMs offer unique opportunities to explain their prediction failures post-hoc. This raises the question of whether LLMs can increase explainability by categorizing reasons for misclassifications.

1.4 The Present Research

Previous research and the identified research gaps ultimately beg the question of what LLMs are able to do in the field of psychological forecasting and where they might face limitations. This study evaluates LLMs' performances in predicting psychological disorder onset through a binary classification problem. The first research question is concerned with comparing the performance of ML models and LLMs to forecast the binary incidence of psychological disorders between two time points, using an exemplary psychological dataset.

Research Question 1: How do traditional ML models and LLMs with prompts perform in terms of forecasting the binary incidence of psychological disorders?

The first research question addresses the *psychological application gap* by applying LLMs in a psychological context using a suitable dataset. It addresses the *comparison and performance gap* by implementing both ML methods and LLMs

with prompts. This allows a first systematic performance comparison between established ML models and LLMs. Finally, it addresses the *implementation gap*. This study uses a variety of prompting strategies and LLMs from different providers to find the best performing method in this field. The second research question investigates the performance of strategies combining ML and LLMs for this task.

Research Question 2: How do unification approaches of ML and LLMs perform in terms of forecasting the binary incidence of psychological disorders?

This research question tackles the *comparison and performance gap* by exploring the performance of ensemble methodologies. These models combine traditional ML strengths and LLM capabilities. We address the *implementation gap* by investigating the optimal application of unification approaches for psychological prediction. The third and last research question explores whether LLMs can identify common failure modes for misclassifications.

Research Question 3: Can post-hoc explanations generated by LLMs for misclassified cases improve the explainability and categorize reasons for failure?

This research question addresses the *explainability gap* by using LLMs to generate explanations for misclassifications. This potentially offers new approaches to understanding prediction failures in the psychological context.

1.4.1 Expected Contributions

This thesis contributes to the theoretical and applied understanding of psychological disorder forecasting with LLMs. It aims to identify the optimal strategy for psychological disorder prediction using LLMs and ML methods. This includes finding the optimal prompting strategies, LLM provider or unification approaches for this task. It is a first approach to establishing performance standards of different LLMs and implementations in the field of psychological disorder prediction. Furthermore, this study intends to develop an explainability framework using LLM-generated failure explanations. This research provides evidence-based guidance for the clinical implementation of future mental healthcare, that could enable personalized treatments, real-time diagnostic monitoring, and improved risk stratification.

Chapter 2

Theoretical Background

This chapter establishes the theoretical foundation for implementing ML models and LLMs. It introduces all methods and forms the basis for the method section. It presents established ML approaches for psychological prediction and prompting strategies for classification tasks using LLMs. It reviews existing literature on unification approaches of ML and LLMs, and failure modes of LLMs.

2.1 ML Classification Algorithms for Psychological Prediction

ML algorithms are the foundation for psychological disorder prediction. These supervised learning methods provide baseline performances. The algorithms selected represent linear, neural, probabilistic, tree-, kernel- and distance-based approaches. Logistic Regression serves as a linear method, modeling disorder probability through linear combinations of predictors (Murphy, 2022). Random Forest implements a tree-based ensemble approach, combining multiple decision trees to capture non-linear relationships and feature interactions while reducing overfitting (Breiman, 2001; Murphy, 2022). Support Vector Classification utilizes a kernel-based method to find optimal decision boundaries in high-dimensional spaces, which is particularly effective for complex psychological classification tasks (Murphy, 2022). Deep Neural Networks employ multi-layer architectures to model complex patterns and interactions within psychological predictors through non-linear transformations (Murphy, 2022). Naive Bayes provides a probabilistic framework based on Bayes' theorem, assuming conditional independence between features while offering computational efficiency (Murphy, 2022). K-Nearest Neighbors implements a distance-based approach, classifying cases based on similar observations in the feature space (Murphy, 2022).

2.2 Prompting Strategies for Classification Tasks Using LLMs

This thesis uses LLMs for binary classification without being explicitly trained on labeled examples for that specific task. Instead, the models use pre-trained knowledge on big datasets and prompting strategies, rather than fine-tuning (Wu, Wang, Wang, & Zheng, 2024). Six different prompting strategies from current literature are employed to evaluate the impact of prompt design on LLMs performances in psychological disorder classification. The analyzed prompts range from a simple prompt, with no added instruction or context, to more complex few-shot and reasoning prompts. All proposed prompting strategies are applied and explained in the field of psychological disorder prediction.

Simple prompt: This prompting strategy represents the baseline prompt. It asks the LLM to determine whether an individual develops a psychological disorder between time points T1 and T2. There was no additional information or instruction provided, solely providing standardized variables (Tavarez-Rodríguez, Sánchez-Vega, Rosales-Pérez, & López-Monroy, 2024).

Class definitions prompt: This prompting strategy provides definitions for YES and NO classifications in the context of psychological disorder classification (Pan, Wu, Li, & Luu, 2024; Tavarez-Rodríguez et al., 2024). These definitions include a specific definition of disorder onset and a list of possible psychological disorders measured with the F-DIPS. This prompting strategy additionally provides the LLM with a general context including the prevalence of psychological disorders.

Profiled simple prompt: This prompting strategy assigns the LLM the role of a clinical psychology expert with specialized knowledge in mental health diagnostics, longitudinal data analysis, diagnostic frameworks, and risk factor analysis (Tavarez-Rodríguez et al., 2024). Additionally, the high-stakes nature of this task and the need for data-driven analysis and methodological rigor is empathized.

Few-shot prompt: This is an in-context learning strategy that provides the LLM with n correctly classified examples prior to the classification task (Brown et al., 2020). For each classification, the examples were chosen as the most similar cases from the training set (Gholamian, Romani, Rudnikowicz, & Skylaki, 2024). This allows the LLM to learn from similar cases before making its own prediction.

Vignette prompt: This prompting strategy transforms the standardized numerical predictors into an opinion-based prompt as a clinical vignette (Haider et al., 2024). A narrative is being created, presenting each classification task as a clinical case narrated and interpreted by a person (Zhou, Zhang, Poon, & Chen, 2023). The individual’s psychological profile and changes over time are presented in natural language with qualitative interpretations. Qualitative interpretations indicate the predictors as above/below average compared to the whole sample and were calculated based on a threshold using standard deviations.

Chain-of-thought prompt: This prompting strategy uses chain-of-thought demonstrations as examples for solving a reasoning task (Wei et al., 2023). It guides the LLM through the classification task by presenting intermediate steps for predicting disorder onset of a person. These include analyzing the demographic variables first, then the baseline predictors at T1, afterwards the changes between T1 and T2 and lastly making a final decision. It additionally includes an demonstration of expert reasoning for an exemplary participant. It asks the LLM to provide its prediction as well as a short explanation.

2.3 Unification Approaches of ML and LLMs

Unification approaches are ensemble methods that combine the strengths of different models and are therefore expected to improve the accuracy of our models. They potentially capture patterns missed by either approach alone. Tavaréz-Rodríguez et al. (2024) proposed three different unification approaches combining LLMs and transformer models. The present research adapts these strategies to unify LLMs and ML models. This section describes ensemble methods following Tavaréz-Rodríguez et al.’s (2024) approaches in the context of psychological disorder forecasting.

Creation of new input: The first unification approach uses LLM-generated predictions as new input for ML models (Tavaréz-Rodríguez et al., 2024). It concatenates the original predictors used for forecasting psychological disorders with LLM responses generated through various prompts for the training data (see Figure 2.1 ¹). It combines quantitative psychological variables with LLM-generated predictions. The new inputs are then used as predictors processed by the best performing ML model.

Proportion of votes: The second unification approach aggregates predictions across multiple models using majority voting (Tavaréz-Rodríguez et al., 2024). The

¹Figures were adapted from Tavaréz-Rodríguez et al. (2024).

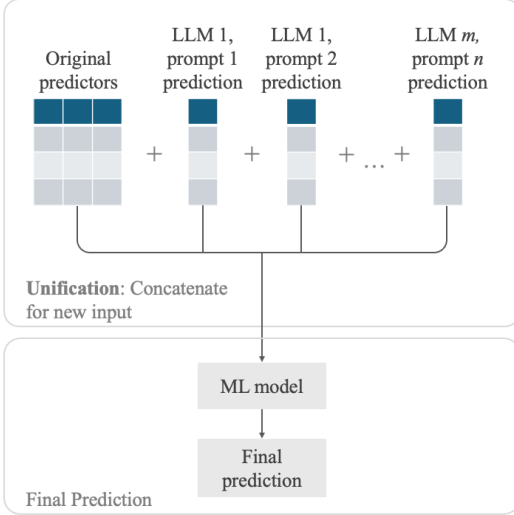


Figure 2.1: Unification approach: Creation of new input for ML model

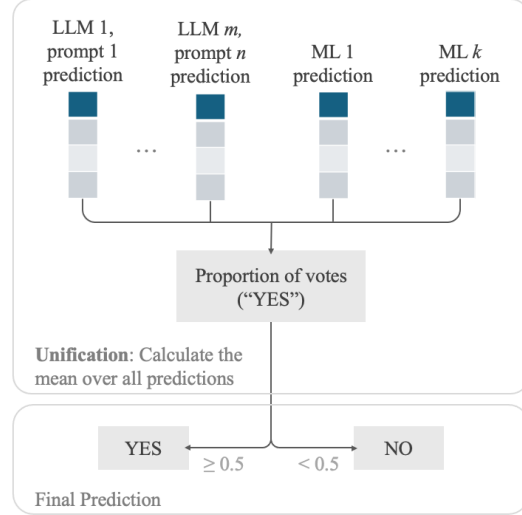


Figure 2.2: Unification approach: Proportion of votes

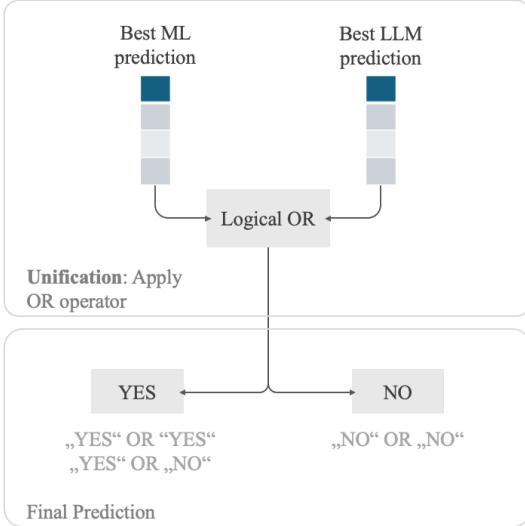


Figure 2.3: Unification approach: Best LLM OR best ML response

proportion of "Yes" votes are being calculated by taking the mean of all selected predictions. This average lies between 0 (all models voted "No") and 1 (all models voted "Yes"). The final classification is "Yes" if the proportion of votes is bigger or equal than a threshold of 0.5 and "NO" otherwise (see Figure 2.2).

Best LLM OR best ML response: The third unification approach combines the best-performing LLM response with the best-performing ML model using a logical OR operation (Tavarez-Rodríguez et al., 2024). If either the ML model or the LLM predicted 1 ("Yes"), the case is ultimately classified as 1 (see Figure 2.3). Only if both models predicted 0 ("NO"), the case is classified as 0.

2.4 LLMs’ Failure Modes: Reasons for Misclassifications

In order to determine the specific reasons for misclassifications regarding the incidence of psychological disorders, we identified several possible reasons from previous research. Based on this literature review, we developed a structured categorization system for LLM misclassifications in psychological prediction. This section examines potential sources of misclassification based on established LLM failure modes and their implementation in psychological diagnosis contexts.

Lack of Context LLMs may struggle when insufficient contextual information is provided (Ray & Bhalani, 2024; Zhou et al., 2023). The model may lack emphasis or indication of the context of the query, such as domain-specific background information about psychological assessments. This can include limited information about the variables, diagnostic criteria, or clinical context of the prediction task.

Lack of Examples LLMs can profit from few-shot learning, however this requires representative examples shown before the real task (Brown et al., 2020; Ray & Bhalani, 2024). Psychological classification might suffer when LLMs lack representative few-shot examples showing correct classifications for similar psychological cases before posing the actual question.

Lack of Counterfactual Demonstrations In addition to correctly specified examples, LLMs might also benefit from exposure to contrasting cases with incorrect classifications (Zhou et al., 2023). In psychological contexts, this involves presenting cases with false psychological diagnoses or conclusions. These misclassifications could improve the model’s robustness against wrong decisions.

Lack of Feedback Another failure mode of LLMs might be the lack of iterative refinement through user feedback (Kim, Lee, Shin, Kim, & Kim, 2024; Ray & Bhalani, 2024). When using interactive chatbots, prediction tasks can require multiple iterations of prompt refinement and dialogues where the user gives feedback. This may not be provided by static prompting approaches.

Lack of Opinion-Based Information Clinical psychology often involves subjective judgment and contextual interpretations. Models may fail when such opinion-based information is missing (Zhou et al., 2023). They might require a reframing of the data as a narrator’s statement and opinions rather than relying solely on quantitative symptoms.

Knowledge Conflicts Psychological diagnostic frameworks evolve continuously, creating potential conflicts between the LLMs memorized training data and current clinical practice (Zhou et al., 2023). In this case, LLMs might rely on outdated memorized facts or conflicting theoretical frameworks, leading to misclassifications based on inconsistent information.

Prediction with Abstention Models may lack sufficient confidence to make a binary decision about psychological disorder incidence (Zhou et al., 2023). This uncertainty can manifest as a misclassifications when the model is being forced to make a decision, even though it lacks confidence and certainty.

Chapter 3

Methods

This thesis evaluated ML algorithms, LLMs, and unification approaches for forecasting psychological disorder incidence between two time points, using an exemplary longitudinal psychological dataset. The present study implemented six ML models, 36 LLM configurations, and 11 unification approaches, in total 53 distinct models. The LLM component implemented six prompting strategies across six state-of-the-art models from leading AI laboratories. To enhance interpretability and inform future prompting strategies, LLMs were asked to provide explanations for misclassifications.

3.1 Data

The present thesis used public available data from the Dresden Predictor Study (DPS)¹, a study investigating predictors of psychological disorders (Trumpf, Margraf, Vriends, Meyer, & Becker, 2010). Detailed information about the procedure, interviews and variables can be found in Trumpf et al.'s (2010) paper.

3.1.1 Participants

The dataset consisted of German women aged 18-25 who were randomly selected from population registers in Dresden, former East Germany, in 1996. Participants completed diagnostic interviews and self-report questionnaires at baseline (T1) and approximately 17 months later (T2). From an initial pool of 5,203 eligible women, 1,538 completed the diagnostic interview at both time points. 309 participants with missing values were excluded from the analysis, providing a final sample size of 1,229 for the present study.

¹Data for: Dresden Predictor Study (DPS) of anxiety disorders and depression in young German Women is publically available at <https://www.psycharchives.org/en/item/dcd1536b-9b99-4ff1-a917-1f9e0528368b>

3.1.2 Measures

The analysis used 22 predictors and one binary outcome. Descriptive statistics of all predictors and the outcome are shown in Appendix A. Further information about the used scales and items can be found in Trumpf et al.'s (2010) paper.

Predictors

The predictor variables included three demographic variables, eleven baseline psychological predictors measured at T1, and eight change predictors capturing the changes in the baseline predictors towards T2. Demographic variables were categorical predictors and consisted of education level, body mass index (BMI), and socioeconomic status (SES). Baseline psychological predictors at T1 included positive mental health, social support, general self-efficacy, life satisfaction, stress, problem-focused coping, emotion-focused coping, anxiety sensitivity, fear of bodily sensations, dysfunctional attitudes, and general psychopathology. Change variables were operationalized as the difference between T1 and T2 and calculated as $T2 - T1$. They captured longitudinal changes in positive mental health, social support, general self-efficacy, life satisfaction, anxiety sensitivity, fear of bodily sensations, dysfunctional attitudes, and general psychopathology.

Target

The target variable was operationalized as the binary incidence of any psychological disorder between T1 and T2. It was coded as 1 if any point prevalence onset was detected during the 17-month follow-up period and 0 otherwise. An incidence only included participants who were healthy at T1 and diagnosed at T2.

Psychological disorders were measured using the "diagnostic interview for psychological disorders" (F-DIPS) (Margraf, Schneider, Soeder, Neumer, & Becker, 1996), a structured interview for the assessment of psychological disorders according to DSM-IV criteria. Disorders of the F-DIPS included anxiety disorders (i.e., panic disorder, agoraphobia, social phobia, specific phobia, generalized anxiety disorder, obsessive-compulsive disorder, PTSD, and acute stress disorder), affective disorders (i.e., dysthymic disorder, major depressive disorder, bipolar disorders, cyclothymic disorder, and mixed anxiety-depressive disorder), somatoform disorders (i.e., hypochondriasis, somatization disorder, conversion disorder, and pain disorder), substance use disorders (i.e., alcohol/medication/drug dependence and abuse) and eating disorders (i.e., anorexia nervosa and bulimia nervosa). The target's distribution was unbalanced. 28.1% of the participants developed a psychological disorder between the two time points, while 71.9% did not.

3.2 Statistical Analyses

All analyses were implemented in Python (Van Rossum & Drake, 2009) (version 3.10). All code was made available open-source on my GitHub² account. In this section, the implementation of all ML models, LLMs, unification approaches and reasons for misclassification are described in detail. See Table 3.1 for an overview of all implemented methods, models and a short description.

3.2.1 Classification Through ML Models

Six ML classification algorithms were implemented, representing different approaches to analyzing psychological data: Logistic Regression, Random Forest, Support Vector Machine, Deep Neural Network, Naive Bayes, and K-Nearest Neighbors. The models were implemented using the package "scikit-learn" (Pedregosa et al., 2011). The dataset was prepared using a stratified 80/20 train-test split. All ML models underwent hyperparameter optimization with a 10-fold cross-validation, were trained on the training data and created predictions for the testing data.

3.2.2 Classification Through LLMs and Prompts

Six LLMs from leading AI laboratories were employed. Since psychological data analysis is a highly complex environment, the models were chosen based on their intelligence and capability to analyze complex relationships. Reasoning models were preferred over fast models and new models over old models. The LLMs consisted of GPT-o3 ("o3-2025-04-16") by OpenAI (OpenAI, 2023), Gemini 2.5 Pro ("gemini-2.5-pro") and Gemma 3 27B ("gemma-3-27b-it") by Google DeepMind (Google DeepMind, 2025a, 2025b), Claude Sonnet 4 ("claude-sonnet-4-20250514") by Anthropic (Anthropic, 2025), DeepSeek Reasoner ("deepseek-reasoner") by DeepSeek AI (DeepSeek AI, 2024), and Grok 3 beta ("grok-3-beta") by xAI (xAI, 2024). Note that the selection of these models was time-bound, given that the models and their availabilities are changing rapidly.

All experiments were conducted between May and July 2025. The models were implemented in Python, connecting to their APIs using the packages "openai" (OpenAI, 2024), "google-genai" (Google, 2024), and "anthropic" (Anthropic, 2024). To evaluate the LLMs' performances and to make them comparable to ML models, all LLMs created predictions solely for the testing data. Each model was given an English (system) instruction and prompt as input. They were implemented using default reasoning effort and temperature, if applicable. For each LLM, we used six different prompting strategies to create predictions for the testing data:

²GitHub: https://github.com/kimmizi/master_thesis/

simple prompt, class definitions prompt, profiled simple prompt, few-shot prompt, vignette prompt and chain-of-thought (CoT) prompt. The used materials for prompting can be found in Appendix B.

All prompts, except for the vignette prompt, were created using a fixed template (e.g., containing context or definitions). Each prompt ended by asking whether the subsequent person developed a psychological disorder between the two time points, followed by a data string with numerical values of all predictors for that participant. In the vignette prompt, qualitative interpretations were created using threshold functions that converted numerical values into qualitative labels. Demographic variables and predictors at T1 were categorized as "above average," "average," or "below average" using ± 0.5 thresholds. Change variables between time points were categorized into seven levels ranging from "strongly decreased" to "strongly increased" using thresholds of ± 0.1 (slight), ± 0.5 (moderate), and ± 1.5 (strong). For the few-shot prompt, the ideal number of examples was evaluated using Gemini and 5, 10, 20, 50, and 100 examples. The best performing few-shot prompt used the 5 most similar examples chosen from the training set. Similarity was determined using the Euclidean distance between feature vectors of the respective participant and participants from the training set, excluding the target variable.

3.2.3 Unification Approaches

Three different unification approaches to unify ML models and LLMs were implemented. For the creation of new input, we used GPT, Gemma, Claude, DeepSeek and Grok to create predictions for the training data. Gemini was not considered because of its long running time. These predictions for the training set were concatenated with the original set of predictors. Similarly for the testing data, the original predictors were concatenated with the LLMs' predictions for the testing data created in Section 3.2.2. The best performing ML model was then trained on the extended training data and created predictions for the extended testing data.

The proportion of votes approach considered all LLMs' predictions for the testing data. The proportion of votes were calculated by taking the mean of predictions and classifying a person ultimately as 1, if the mean is bigger or equal to 0.5 and 0 otherwise. Proportion of votes were calculated using the following (sub)groups of models: all models, all ML models, all LLMs, all simple prompts, all class definitions prompts, all profiled simple prompts, all few-shot prompts, all vignette prompts and all CoT prompt. The last unification approach was implemented using an OR operator on the best LLM response and the best ML model. A person was ultimately classified as 1 if either the ML model or the LLM predicted 1, and as 0 if both models predicted 0.

Table 3.1: Overview of all methods, models and descriptions for research questions (RQ) 1 to 3.

Method	Models	Description
Classification through ML models (RQ1)	Logistic Regression, Random Forest, Support Vector Machine, Deep Neural Network, Naive Bayes, K-Nearest Neighbors	Supervised learning algorithms used for binary classification with cross-validation and hyperparameter tuning.
Classification through LLMs and prompts (RQ1)	GPT-o3 (OpenAI), Gemini 2.5 Pro (Google DeepMind), Gemma 3 27B (Google DeepMind), Claude Sonnet 4 (Anthropic), DeepSeek Reasoner (DeepSeek AI), Grok 3 beta (xAI)	Simple prompt: No added instructions or context. Class definitions prompt: Explicit definitions of YES and NO. Profiled simple prompt: Allocating the role of a clinical psychologist and expert in analyzing data. Few-shot prompt: Top-5 similar examples as in-context learning before classification. Vignette prompt: Clinical vignette with qualitative interpretations. Chain-of-thought prompt: Intermediate reasoning steps and chain of thought demonstration.
Unification of knowledge (RQ2)	Ensemble methods using ML models and LLMs	Creation of new input: Concatenating predictors and the responses by the LLMs and different prompts as new input in a ML model for classification. Proportion of votes: Calculating the proportion of YES and NO answers from all models: A threshold of 0.5 is used to ultimately classify a person. Best LLM OR best ML response: Classification based on a logical OR operation of the responses of the best LLM and the best ML model.
Reasons for Misclassification (RQ3)	All LLMs were prompted to categorize misclassified cases into common failure modes.	Possible reasons: Lack of context, lack of examples, lack of counterfactual demonstrations, lack of feedback, lack of opinion-based information, knowledge conflicts, and prediction with abstention.

3.2.4 Reasons for Misclassifications

In order to ask for reasons of misclassification, all LLMs were post-hoc presented with information about their misclassified cases and instructed to categorize the reason of failure. For each misclassified case, the model was given the respective prompt, its incorrect response, the correct response, and if applicable with its reasoning or thinking process. LLMs were instructed to categorize why they misclassified the data in order to improve the accuracy of predicting the incidence of a mental disorder between two time points. Then they were asked to choose one or many of the provided categories and answer with the most important reason first. The categories consisted of: lack of context, lack of examples, lack of counterfactual demonstrations, lack of feedback, lack of opinion-based information, knowledge conflicts, and prediction with abstention. The used materials for prompting can be found in Appendix C.

3.3 Evaluation Criteria

Model performance was measured by comparing the predictions with the ground truth. We evaluated all models based on accuracy (proportion of correct predictions), Matthew’s Correlation Coefficient (MCC, which accounts for class imbalance), sensitivity (true positive rate), specificity (true negative rate), and macro F1 score (harmonic mean of precision and recall averaged across classes). In clinical predictions, it is important to consider both global metrics (accuracy, MCC, F1) as well as sensitivity and specificity. The sensitivity and specificity are important for clinical implications and psychological disorder detection.

Chapter 4

Results

4.1 Comparing ML Models and LLMs for Forecasting Psychological Disorder Incidence

The first research question was concerned with how the performance of ML models and classification through LLMs and prompts compare in forecasting the binary incidence of psychological disorders.

4.1.1 Performance of ML Models

All ML models achieved good accuracy, ranging between 0.71 and 0.74 (Table 4.1). The ML models differed in their ability to correctly detect true positive cases (i.e., detect psychological disorder onset). While the KNN model was not able to perform better than chance ($MCC = -0.01$), the highest MCC was achieved by the Naive Bayes model ($MCC = 0.24$). The Support Vector Machine was able to identify all negative cases correctly (specificity = 1.00), but it was not able to detect the majority of positive cases (sensitivity = 0.04). DNN and Naive Bayes showed the highest capability of correctly detecting psychological disorder onset, both achieving a sensitivity of 0.22, while maintaining a high specificity of 0.92 and 0.94, respectively. Overall, the Naive Bayes model was the best performing ML model, achieving the highest accuracy, MCC, sensitivity, and macro F1 score.

4.1.2 Average Performance of LLMs

Averaging over the six different prompting strategies, GPT and Grok evolved as the best performing LLMs (Table 4.2). GPT showed on average the highest accuracy of 0.68. Grok achieved on average the highest macro F1 score of 0.56. The highest average MCC of 0.11 was achieved by GPT and Grok. GPT seemed to be able

Table 4.1: Performance of all ML models.

ML Model	Acc.	MCC	Sens.	Spec.	F1
Logistic Regression	0.73	0.20	0.20	0.93	0.57
Random Forest	0.73	0.20	0.20	0.93	0.57
Support Vector Classification	0.73	0.18	0.04	1.00	0.52
Deep Neural Network	0.72	0.18	0.22	0.92	0.57
Naive Bayes	0.74	0.24	0.22	0.94	0.58
K-Nearest Neighbors	0.71	-0.01	0.01	0.98	0.50

Note. Acc. = Accuracy, MCC = Matthews Correlation Coefficient, Sens. = Sensitivity (Recall), Spec. = Specificity, F1 = Macro F1 score. The highest values for each metric are shown in bold.

to detect negative cases best (specificity = 0.86) while having a low sensitivity of 0.23. Grok was able to detect positive cases best (sensitivity = 0.65), while having a moderate specificity of 0.46.

Table 4.2: Average performance across all prompt types for each LLM.

LLM	Acc.	MCC	Sens.	Spec.	F1
GPT	0.68	0.11	0.23	0.86	0.55
Gemini	0.62	0.08	0.38	0.71	0.54
Gemma	0.46	0.03	0.64	0.39	0.51
Claude	0.57	0.02	0.37	0.65	0.51
DeepSeek	0.64	0.07	0.27	0.78	0.53
Grok	0.51	0.11	0.65	0.46	0.56

Note. Acc. = Accuracy, MCC = Matthews Correlation Coefficient, Sens. = Sensitivity (Recall), Spec. = Specificity, F1 = Macro F1 score. The highest values for each metric are shown in bold.

4.1.3 Average Performance of Prompting Strategies

Considering the different prompting strategies, the profiled simple prompt, the few-shot prompt and the vignette prompt achieved the highest performance when averaged over all implemented LLMs (Table 4.3). The best averaged accuracy of 0.64 was achieved by the vignette prompt. This prompting strategy showed the highest capability on average to detect negative cases correctly (specificity = 0.75). The highest MCC of 0.12 and highest macro F1 score of 0.56 were achieved by the few-shot prompt. It showed on average a moderate sensitivity of 0.51 and a moderate specificity of 0.62, suggesting that it is able to balance the detection of both positive and negative cases. The best capability to detect positive cases was achieved by the profiled simple prompt (sensitivity = 0.53).

Although the difference between the average metrics is minor, the simple prompt, the class definition prompt and the chain-of-thought prompt performed worse on average. The chain-of-thought prompt achieved the lowest MCC of all prompts and performed on average not better than random (MCC = 0.00).

Table 4.3: Average performance across all LLMs for each prompt type.

Prompting strategy	Acc.	MCC	Sens.	Spec.	F1
Simple prompt	0.56	0.08	0.49	0.58	0.54
Class definitions prompt	0.59	0.02	0.32	0.70	0.51
Profiled simple prompt	0.56	0.11	0.53	0.57	0.55
Few-shot prompt	0.59	0.12	0.51	0.62	0.56
Vignette prompt	0.64	0.09	0.33	0.75	0.54
Chain-of-thought prompt	0.55	0.00	0.37	0.62	0.50

Note. Acc. = Accuracy, MCC = Matthews Correlation Coefficient, Sens. = Sensitivity (Recall), Spec. = Specificity, F1 = Macro F1 score. The highest values for each metric are shown in bold.

4.1.4 Detailed Performance of LLMs and Prompting Strategies

The performance of LLMs for forecasting psychological disorder onset depended on both the specific LLM and the used prompting strategy. The results for all combinations of the LLMs and prompting strategies can be found in Appendix D. The most effective prompting strategies varied depending on the specific LLM. For the simple prompt, GPT performed the best and achieved an accuracy of 0.70 and MCC of 0.14. It showed a low sensitivity of 0.23, but the highest specificity of 0.88 out of all models. The best sensitivity achieved Grok for the simple prompt (sensitivity = 0.93), detecting almost all positive cases. However, Grok achieved the lowest accuracy out of all models of 0.36 for the simple prompt.

For the class definitions prompt, GPT again achieved the best accuracy of 0.70 and specificity of 0.92. The best macro F1 score of 0.54 was achieved by Gemini. The highest MCC of 0.10 and the highest sensitivity of 0.62 were achieved by Grok. Gemma and Claude scored for the class definitions prompt a negative MCC of -0.05 and -0.07, respectively. These models performed worse than by chance when implementing the class definitions prompt.

For the profiled simple prompt, GPT and DeepSeek achieved the highest accuracy of 0.72. The best MCC over all LLMs and all prompts was achieved by GPT and the profiled simple prompt (MCC = 0.18.). While Gemma achieved a perfect sensitivity of 1.00 and detecting all positive cases, it was not able to detect

any healthy participants (specificity = 0.00). Gemma’s MCC of 0.00 and accuracy of 0.28 conveys that it predicted 1 ("Yes") for all participants and was not able to capture any meaningful pattern. DeepSeek achieved the highest specificity of 0.95 for the profiled simple prompt. The best macro F1 of 0.59 was achieved by Grok.

For the few-shot prompt, DeepSeek and Grok performed best with DeepSeek having the highest accuracy of 0.64 and Grok having the highest MCC of 0.17. While DeepSeek was able to detect negative cases best (specificity = 0.74), Grok performed best in detecting the positive cases (sensitivity = 0.70). For the vignette prompt, Gemma achieved the highest accuracy of 0.67 and the best specificity of 0.84. Grok achieved the best MCC of 0.15, the best sensitivity of 0.51 and the best macro F1 score of 0.58.

For the chain-of-thought prompt, GPT performed best in all metrics except for sensitivity. While Gemma had the best sensitivity out of all models (sensitivity = 0.87), GPT scored the lowest (sensitivity = 0.12). Additionally, Gemini achieved next to GPT the best macro F1 score of 0.52. Gemma, Claude and DeepSeek performed worse than chance, all having a negative MCC ranging -0.02 to -0.06.

Out of all the combinations of six LLMs and six prompting strategies, GPT using the profiled simple prompt performed best in terms of accuracy and MCC. The highest sensitivity was achieved using Gemma and the profiled simple prompt, even though not capturing any meaningful patterns. The highest specificity overall was achieved by implementing DeepSeek and the profiled simple prompt. The best macro F1 score was achieved using Grok and the few-shot prompt.

The results show that the performance of different prompts varied depending on the selected LLM. The profiled simple prompt seems to be working well to detect positive cases. The vignette prompt showed the most constant results for the best accuracy and specificity. The few-shot prompt showed the best results in terms of MCC and macro F1 score, balancing best the detection of both positive and negative cases.

4.1.5 Performance Patterns and Model Comparisons

Figure 4.1 plots the MCC and accuracy for all ML models, LLMs and prompts. None of the models achieved a higher accuracy than 0.8 or a higher MCC than 0.25. Naive Bayes achieved the highest performance out of all models. The ML models, except for KNN, outperformed most of the LLMs. However, the profiled simple prompt and the few-shot prompt (e.g., implemented by GPT, Grok or DeepSeek) showed performances close to the performance of ML models.

Figure 4.2 shows an overview of all ML models and LLMs with respect to their achieved sensitivity and specificity. The scatter plot shows that ML models

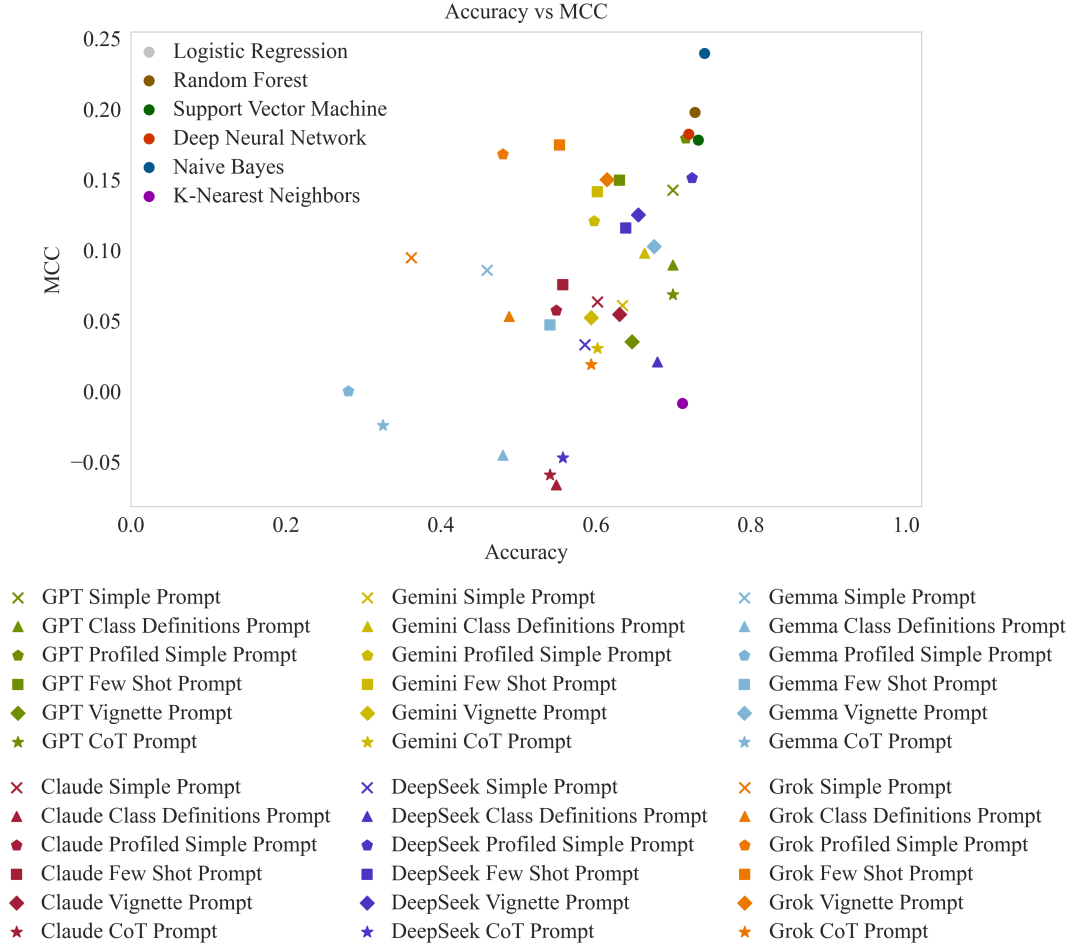


Figure 4.1: MCC vs Accuracy for all ML models and LLMs

Note. The performance of the Logistic Regression model is exactly equal to the performance of the Random Forest model and not displayed in this plot.

and LLMs lie on a diagonal between high specificity/low sensitivity, and low specificity/high sensitivity. Most models fell into two categories: those that correctly identified healthy participants but missed disorder cases (upper left quadrant), and those that detected disorder cases well but incorrectly classified many healthy participants as having disorders (lower right quadrant). Some models balanced both sensitivity and specificity, but without achieving very high values for either metric. The lower left quadrant (models that perform poorly on both measures) and the upper right quadrant (models that excel at both measures) remained largely empty.

All ML models showed relatively high specificity but low sensitivity. This means ML models correctly identified healthy participants but frequently missed those who developed disorders. LLMs, on the other hand, showed a much more diverse and balanced distribution across the entire range of sensitivity and specificity. In extreme cases, the KNN model predicted "No disorder onset" (0) and the Gemma

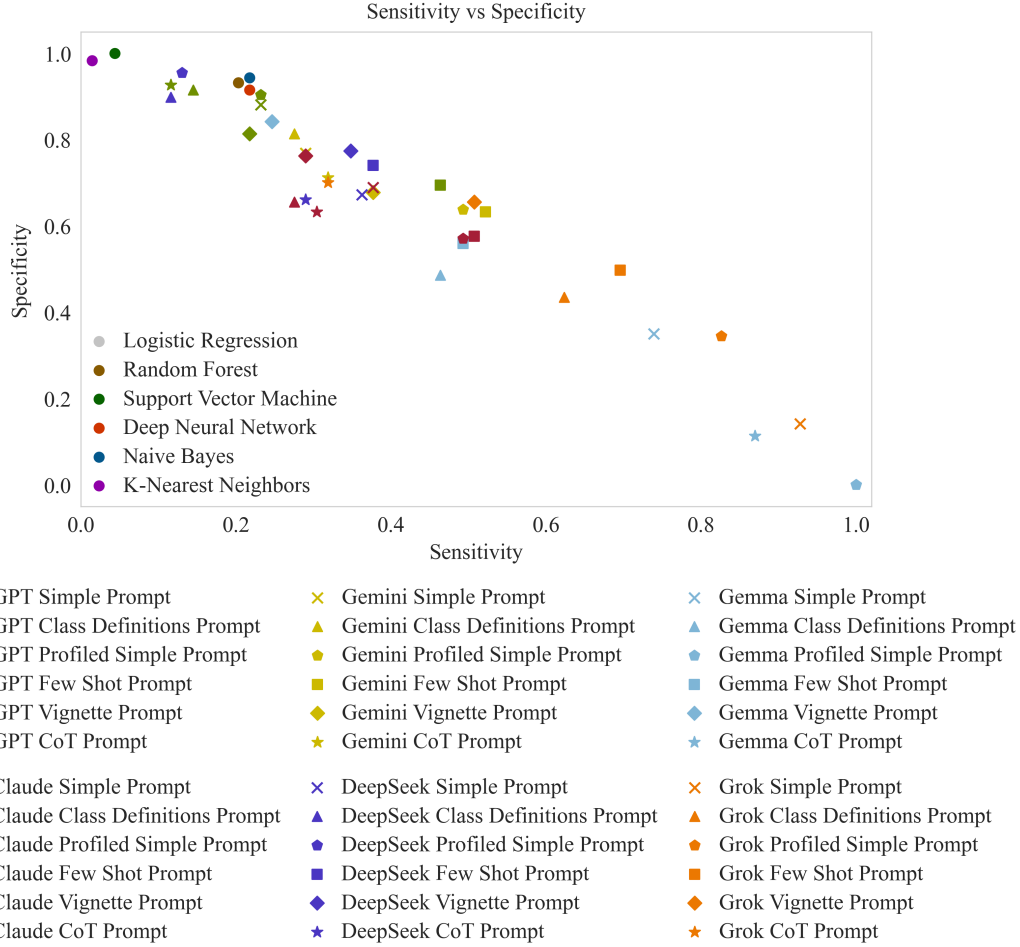


Figure 4.2: Sensitivity vs Specificity for all ML models and LLMs

Note. The performance of the Logistic Regression model is exactly equal to the performance of the Random Forest model and not displayed in this plot.

profiled simple prompt predicted "Disorder onset" (1) for almost all participants. These models did not capture any meaningful pattern, despite their achieved accuracies (KNN: Accuracy = 0.71 and Gemma profiled simple: Accuracy = 0.28).

To conclude research question 1, ML models outperformed LLMs in terms of accuracy and MCC, with Naive Bayes achieving the best performance. However, ML models showed a low sensitivity and were not able to correctly forecast the majority of psychological disorder onsets. While LLMs were in general not able to predict the correct class in as many cases as the ML models, they showed a more balanced sensitivity and specificity. Among LLMs, GPT and Grok performed best on average, with GPT showing the highest specificity and Grok best sensitivity. The profiled simple prompt, few-shot prompt and vignette prompt emerged as the most effective strategies. The best LLM implementation used GPT and a profiled simple prompt. The results reveal a trade-off between specificity and sensitivity, with none of the models excelling at both metrics.

4.2 Unification Approaches of ML and LLMs for Forecasting Psychological Disorder Incidence

The second research question was concerned with how unification approaches of ML and LLMs perform in forecasting the binary incidence of psychological disorders. Table 4.4 shows the metrics of all unification approaches. The creation of new input that concatenated all the LLM predictions (except for Gemini) with the original predictors was used as input for the best-performing ML model: the Naive Bayes model. It achieved the highest accuracy of 0.74 and MCC of 0.25 out of all unification approaches. It showed very high ability to forecast healthy participants (specificity = 0.93), but relatively low ability to detect psychological disorders (sensitivity = 0.26).

Table 4.4: Performance of all unification approaches.

Unification Approach	Unified models	Acc.	MCC	Sens.	Spec.	F1
Creation of new input	Naive Bayes & all LLM predictions	0.74	0.24	0.23	0.94	0.58
Proportion of votes	All models	0.65	0.07	0.26	0.80	0.53
	ML models	0.72	0.10	0.07	0.97	0.52
	LLM models	0.61	0.03	0.29	0.74	0.51
	Simple prompts	0.61	0.04	0.32	0.72	0.52
	Class def. prompts	0.64	0.03	0.22	0.81	0.51
	Profiled s. prompts	0.64	0.17	0.46	0.71	0.59
	Few-shot prompts	0.62	0.18	0.54	0.66	0.60
	Vignette prompts	0.65	0.07	0.25	0.81	0.53
	CoT prompts	0.59	-0.02	0.26	0.72	0.49
Best ML OR best LLM	GPT profiled simple prompt OR Naive Bayes	0.72	0.23	0.35	0.86	0.60

Note. Acc. = Accuracy, MCC = Matthews Correlation Coefficient, Sens. = Sensitivity (Recall), Spec. = Specificity, F1 = Macro F1 score, Class def. prompts = Class definitions prompts, Profiled s. prompts = Profiled simple prompts. The highest values for each metric are shown in bold.

Out of all proportion of votes (PoV) approaches, the PoV using only ML models achieved the highest accuracy of 0.72 and highest specificity of 0.97. The best MCC of 0.18, sensitivity of 0.54 and macro F1 of 0.60 was achieved by PoV using only few-shot prompts. While the PoV that used profiled simple prompts also performed well, the PoV that used chain-of-thought prompts scored the lowest performance, predicting slightly worse than chance (MCC = -0.02).

The unification approach using the logical OR operator between the best LLM (GPT and profiled simple prompt) and the best ML model (Naive Bayes) achieved overall solid performance. It achieved high accuracy of 0.72, high MCC of 0.23, moderate sensitivity of 0.35, high specificity of 0.86, and high macro F1 of 0.60. Although, it increased performance compared to using the GPT profiled simple prompt alone, it decreased performance compared to using Naive Bayes alone.

To conclude research question 2, most unification approaches scored good performances. The creation of new input performed best, even performing a bit better than the best ML model, Naive Bayes when trained on the original predictors alone. This means that including LLM predictions in the training set as predictors increased the performance of ML models, even though only insignificantly. However, the remaining unification approaches were not able to perform better than the best performing ML models. Still, they were mostly able to score a more robust performance compared to the LLMs' performance.

4.3 Reasons for Misclassifications by LLMs

The third research question was concerned with the question whether explanations generated by LLMs for misclassified cases can improve the explainability of predictions for psychological disorder onset. LLMs were able to categorize reasons for failure. Lack of context was the most frequent reason for misclassification across all LLMs and prompting strategies, mentioned in 58.3% of cases. The next most frequent categories were the lack of opinion-based information (28.5%) and lack of counterfactual demonstrations (22.1%). Lack of examples were mentioned in 16.1% of the cases and knowledge conflicts in 12.7%. Finally, prediction with abstention was mentioned in only 3.1% of the misclassified cases and lack of feedback was not mentioned as a reason for misclassification (0.00%) by any of the models.

The different LLMs provided different patterns for reasons of misclassifications. GPT predominantly mentioned lack of context across all prompting strategies (Figure 4.3¹). It showed minimal variation in reasons, with lack of examples emerging as the second most common reason, mostly for the few-shot, profiled simple and class definitions prompts. GPT rarely mentioned knowledge conflicts or prediction with abstention.

Gemini exhibited a very diverse pattern for reasons for misclassifications (Figure 4.4). It frequently mentioned knowledge conflicts, particularly for the profiled

¹*Note.* Numbers in brackets indicate the absolute number of misclassified cases. Lack of countf. ex = Lack of counterfactual examples, Pred. with abstention = Prediction with abstention, Lack of opinionb. info. = Lack of opinion-based information, Class def. prompt = Class definitions prompt, Profiled s. prompt = Profiled simple prompt.

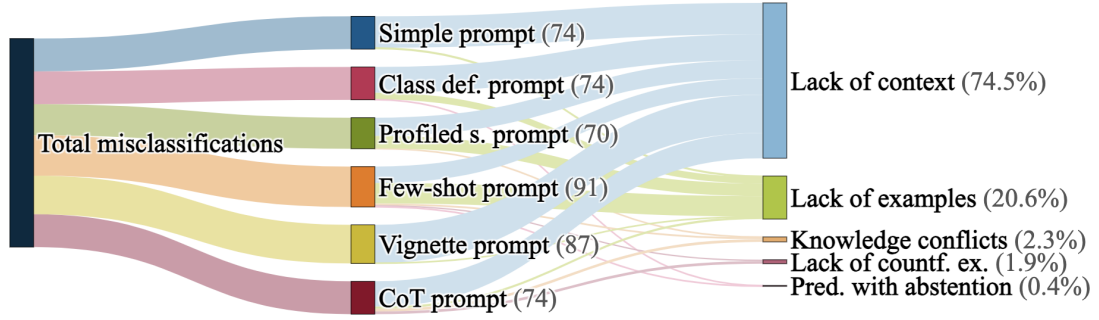


Figure 4.3: GPT: Reasons for misclassifications

simple, vignette, and CoT prompts. Furthermore, it mentioned lack of opinion-based information and lack of context across all prompting strategies. Other reasons for failure that were mentioned by all prompting strategies included lack of (counterfactual) examples and prediction with abstention.

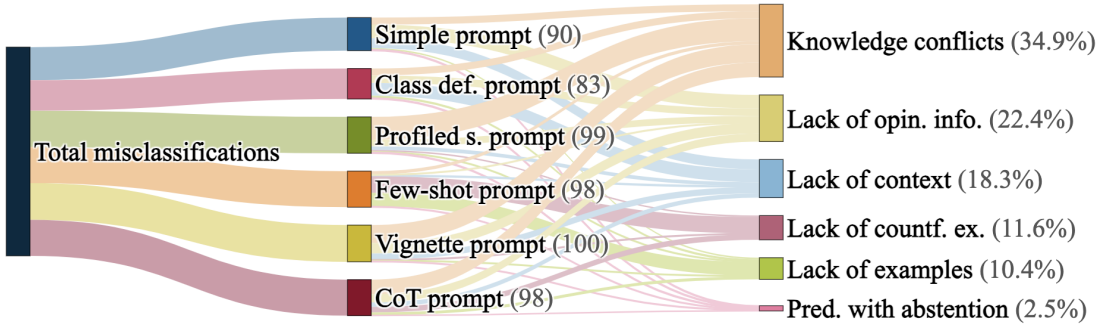


Figure 4.4: Gemini: Reasons for misclassifications

Gemma showed the most consistent pattern, attributing misclassifications to lack of context in most cases, regardless of the prompting strategy used (Figure 4.5). For few-shot and profiled simple prompts, it mentioned lack of examples and knowledge conflicts, respectively. It did not mention other failure modes.

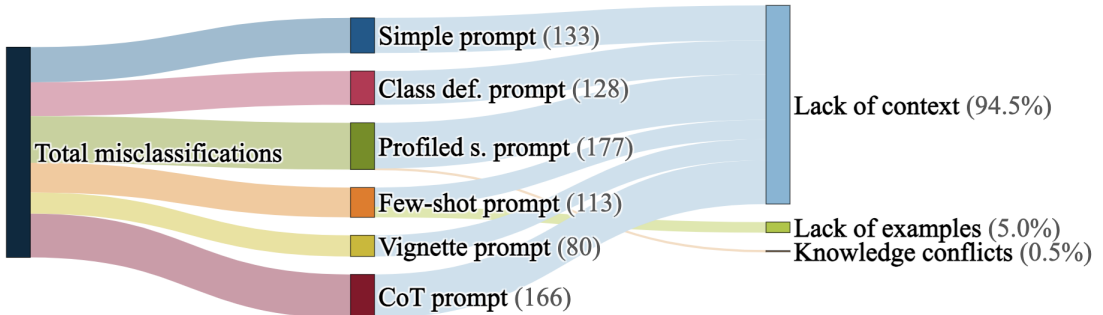


Figure 4.5: Gemma: Reasons for misclassifications

Claude displayed an attribution pattern dependent on the chosen prompting strategy (Figure 4.6). While emphasizing lack of context in the simple, class

definitions, vignette, and CoT prompts, it mostly shifted toward lack of examples and lack of counterfactual demonstrations in the profiled simple and few-shot prompts. Other reasons were mentioned only rarely.

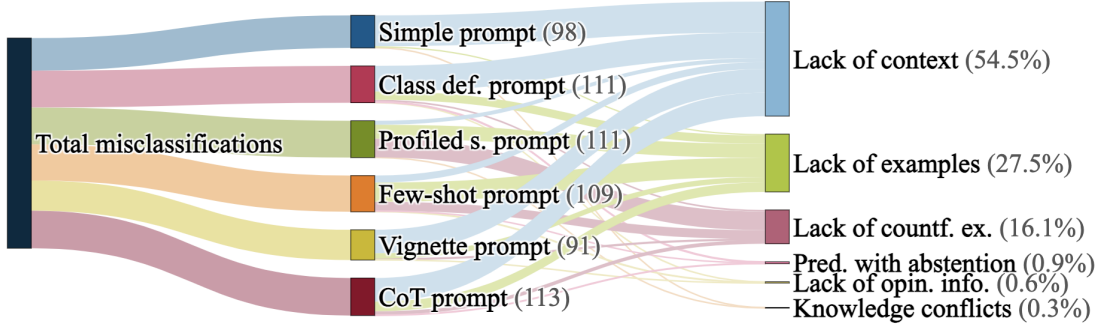


Figure 4.6: Claude: Reasons for misclassifications

DeepSeek stated lack of opinion-based information as the primary reason for misclassifications across most prompting strategies (Figure 4.7). In the few-shot prompt, lack of counterfactual demonstrations was referred to most. Other reasons for misclassifications include lack of context, especially for the class definitions prompt, lack of examples, for the few-shot prompt, knowledge conflicts and prediction with abstention.

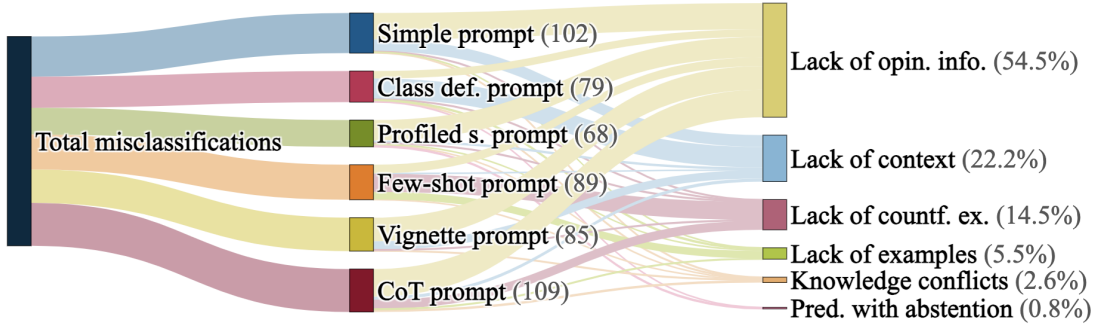


Figure 4.7: DeepSeek: Reasons for misclassifications

Grok also showed a pattern of reasons dependent on the chosen prompting strategy (Figure 4.8). For the simple prompt, lack of context was the only reason for misclassifications. For the class definitions, profiled simple and vignette prompts, lack of context, lack of opinion-based information and lack of examples were the most important reasons. For the few-shot and CoT prompts, lack of counterfactual examples were stated most frequently.

To conclude research question 3, LLMs were able to categorize reasons for failure, even though the reliability of such explanations may have varied across models and prompting strategies. While LLMs differ in their pattern of reasons for misclassifications, lack of context emerged as the most important reason. GPT,

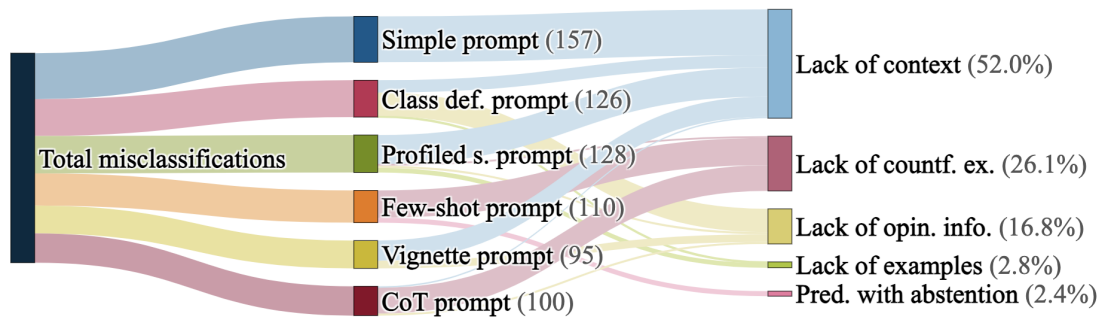


Figure 4.8: Grok: Reasons for misclassifications

Gemma, Claude and Grok mentioned lack of context, Gemini knowledge conflicts and DeepSeek lack of opinion-based info most frequently as the primary reason for misclassifications. Simple prompts most often generated lack of context as reason for misclassification, while few-shot prompts generated lack of examples or counterfactual demonstrations. There is a predominance of lack of context, information or examples over knowledge conflicts and prediction with abstention.

Chapter 5

Discussion

The present thesis investigated whether LLMs are able to forecast the binary incidence of psychological disorder onsets. It compared ML algorithms, LLMs, and unification approaches for forecasting psychological disorder incidence between two time points. It used LLMs to categorize misclassified cases into common failure modes of LLMs and explain reasons of misclassifications.

5.1 ML and LLM Performance: Accuracy-Sensitivity Trade-off

The findings of the first research question show that LLMs were able to forecast psychological disorder onset between two time points. LLMs scored accuracies ranging between 0.28 and 0.72 with mean accuracies over different prompting strategies of 0.55 to 0.64. These results were relatively consistent using a variety of LLMs and different prompting strategies, suggesting that our findings are valid. Additionally, the results align with previous performances of LLMs. Caruccio et al. (2024) investigated ChatGPT’s performance for providing diagnoses using medical data and scored accuracies between 0.01 and 0.67. However, there is still room for performance improvement, when comparing these results to previous studies using LLMs to classify other medical outcomes. Dennstädt et al. (2024) used LLMs for oncological trial classification tasks and found accuracies ranging from 0.86 to over 0.95. Haider et al. (2024) used GPT and Gemini to classify breast conditions and scored accuracies between 0.71 and 0.98.

However, LLMs did not outperform most ML methods in terms of their accuracy and MCC. The ML models achieved accuracies between 0.71 and 0.74. These results are aligned with previous studies predicting psychological outcomes, such as outcomes of treatment with mean accuracies ranging 0.56 to 0.75 (Sajjadian et al., 2021). The ML models scored sensitivities ranging between 0.01 and 0.22

and specificities between 0.92 and 1.00. This discrepancy is most likely due to the training on unbalanced data (28.1% out of all participants developed a psychological disorder between the two time points). In comparison to a study predicting recovery from anxiety disorders within a balanced setting (54.6% patients recovered), this study scored sensitivities between 0.62 and 0.65 and specificities of about 0.63 (Bokma et al., 2022). This highlights the challenge that unbalanced data pose for traditional prediction models, such as ML. Similar to Alhindi, Muresan, and Nakov (2024), future studies might explore the use of LLMs to handle the imbalance of data. Creating synthetic data with LLMs can be a promising approach used to create more balanced data in the context of psychological disorder prediction.

While ML models lacked high sensitivity and were not able to correctly forecast the majority of psychological disorder onsets, LLMs often showed a more balanced sensitivity and specificity. However, the best performing LLM (GPT profiled simple prompt) only detected 23% out of all participants who actually developed a psychological disorder. The remaining 77% were missed (false negatives). This indicates poor performance at detecting true cases of disorder onset - the model failed to identify most people who actually develop disorders. Other models with a higher sensitivity, for example the Grok few-shot prompt (accuracy = 0.55), detected 70% of actual psychological disorder onsets as positive cases, but classified only 50% of actual healthy participants as negative cases. In comparison, Caruccio et al. (2024) found sensitivities between 0.01 and 0.67, again validating our results. Ultimately, the models were not able to simultaneously achieve a high sensitivity and a high specificity. There seems to be a trade-off between the two metrics. This trade-off might be partially explained by the unbalanced nature of psychological disorders. There might be a prior bias of LLMs toward predicting "no disorder", inflating specificity at the expense of sensitivity. In the context of clinical predictions, it is crucial to have accurate predictions as well as a high sensitivity and specificity. Still, the sensitivities and specificities of our LLMs are comparable to findings from other studies, while the ML models lack high sensitivity.

5.2 Unification Approaches and Model Selection Guidelines

The results from research question 2 show that despite most unification approaches scoring good performance, they were mostly not able to outperform the best ML models. Only the creation of new input using LLMs improved the performance of ML models, although only insignificantly. At the same time, unification approaches scored much more stable and robust results than ML models or LLMs alone.

Including LLMs predictions as new input for ML models might increase performance and sensitivity. Proportion of votes averages several predictions and is less sensitive to single incorrect predictions. This soft approach to classification achieved good performance, even though not best-performing, and more balanced and stable metrics. Using an OR operator on the best performing ML and LLM prediction increased sensitivity, even though it potentially increased the number of positive cases without justification.

In conclusion, the choice of using ML models, LLMs or unification approaches for psychological prediction seems to be dependent on the specific context. ML models have proven to be fast and have a high performance (Caruccio et al., 2024). However, they require big datasets of high quality and are only as good as their training data. LLMs are pre-trained on large sets of data from many different fields. LLMs are able to analyze symptoms and create intelligent diagnoses (Caruccio et al., 2024). However, their NLP background implies that their response is dependent on the specific prompt formulation. Small prompt alternations might affect the results in severe ways and reduce reliability of responses (Caruccio et al., 2024; Reiss, 2023). Still, LLMs show the capability to overcome some limitations of traditional ML. In the context of this study, LLMs don't require additional training, they don't overfit and are less affected by unbalanced data. However, their responses can be "ambiguous and out of context, so providing incorrect diagnoses" (Caruccio et al., 2024), they might struggle with precise diagnoses.

ML models pose a strong baseline for psychological prediction models. They should be implemented in cases where a lot of high-quality data is available, there is time to properly train ML models or feature importance is necessary. However, in psychological research, acquiring large datasets is often problematic and it might be difficult to measure latent constructs invariantly. LLMs are a promising approach for psychological disorder forecasting when only small datasets are available, training is challenging or real-time classifications are needed, due to their fast provision of predictions. While in low-stake contexts it might be sufficient to use ML models or LLMs alone, in high-stake context it might be appropriate to implement unification approaches due to their robust results. However, unification methods are the most time-consuming and complex approaches, with performances not necessarily increasing.

5.3 Enhancing Explainability with Reasons for Misclassifications

The results from research question 3 show that using LLMs to state reasons for failure is a promising approach for prompt refinement and understanding LLMs rationalities. LLMs were able to categorize reasons for failure, even though the consistency of explanations varied across models and prompting strategies. There is a predominance of lack of context, information or examples over knowledge conflicts and prediction with abstention. This suggests that performance improvements can be achieved in practice through better prompt design and data provision rather than retraining the model or making architectural modifications.

The different patterns in categorizing reasons for misclassifications additionally strengthen our results for different performances of LLMs. For some of the models showing low performance metrics, inconsistencies were also found for their reasons of misclassifications. In this sense, Gemma was not able to differentiate between different reasons of misclassifications and mentioned for most cases the lack of context, independent if actually more context was given in enhanced prompting strategies. Gemini and DeepSeek showed a more diverse pattern of reasons, however without exhibiting a clear relationship between prompting strategies and reasons. The two best performing models, GPT and Grok, as well as Claude, exhibited a more clear pattern, attributing most reasons dependent on the chosen prompt. This suggests that models that performed better in predicting psychological disorder onset could have the tendency to meaningfully attribute reasons for failure.

Future studies might focus on enhancing prompt engineering using these results. Asking a LLM in real time why a prediction was misclassified could improve the prediction. This way, prompts could be interactively refined. However, the finding that lack of feedback was never mentioned from any of the models might suggest that interactive refinement of prompts might be not as effective as promised.

5.4 Limitations and Future Studies

The present thesis has a number of limitations that should be addressed in future work. One limitation concerns the choice of data. The present study used an exemplary dataset consisting of young, German women, aged 18-25 (Trumpf et al., 2010). Additionally, the data was collected between 1996 and 1999, making it possibly outdated today. From today's perspective, the F-DIPS questionnaire has a less differentiated concept of psychological disorders, potentially resulting in less reliable and comparable results. Analyses using the F-DIPS covered psychological

disorders including anxiety disorders, affective disorders, somatoform disorders, substance use disorders and eating disorders. However, it misses for example neurodevelopmental disorders (e.g., autism spectrum disorder), psychotic spectrum disorders (e.g., schizophrenia), personality disorders (e.g., borderline disorder) and many more. Additionally, participants from 25 years ago may have been more affected by the perceived stigma of psychological disorders than participants today. All these factors might have added to missing psychological disorders in participants, decreasing the quality of the dataset and reducing the generalizability of this thesis’ results. Future studies could reproduce our findings using modern, up-to-date data with a large, more diverse sample. Additionally, this study used a dataset made publicly available by the Leibniz Institute for Psychology (ZPID), which is why we cannot rule out the possibility that some of the LLMs were trained using this open-source data.

The choice of ML models, LLMs, prompting strategies and unification approaches in this study might be arbitrary. Especially LLMs from new, small companies or laboratories or other advanced prompting strategies might have not been implemented in this study, possibly limiting the universality of the results. Future studies might also compare (fine-tuned) transformer models, such as BERT or Llama, or Low-Rank Adaptation of LLMs. They might also explore prompting strategies, such as including explanations of relationships between predictors and the target. Another approach for future studies could be to extract the reasoning of an LLM and use second LLM to create predictions. Unification approaches could also include weighted linear combinations of different ML and LLMs predictions or combining ML models’ feature importance with LLM reasoning capabilities.

5.5 Capabilities and Limitations of LLMs in Psychological Disorder Forecasting

This thesis ultimately begs the question of what are LLMs’ capabilities for forecasting psychological disorders and where they face limitations. Sartori and Orrù (2023) emphasize that the identification of shortcomings and not realizable tasks of LLMs is important for future improvement of LLMs. In contrast, well functioning LLM approaches hold a lot of potential and clinical implications in the field of psychology. The central question of this section is whether LLMs should ultimately be used for forecasting psychological disorders.

LLMs are able to tackle many reasoning and problem-solving tasks from cognitive psychology (Binz & Schulz, 2023; Sartori & Orrù, 2023). They are able to examine psychological texts, detect psychological constructs, and analyze psy-

chootherapy sessions (Abdurahman et al., 2024; Rathje et al., 2024; Stadel et al., 2024). In addition to that, this study shows that LLMs are able to forecast the binary incidence of psychological disorders between two time points.

Despite the great potential LLMs hold for predicting psychological disorder onset, there are several aspects to consider that might limit LLMs' practical applications. For instance, LLMs were shown to reproduce human biases, such as primacy, recency or framing effects as well as representativeness and availability heuristics (Sartori & Orrù, 2023). Furthermore, LLMs are also known to produce non-human biases called "hallucinations" (Ji et al., 2023), incorrect, nonsensical, or inconsistent responses produced by limited context. Since predicting psychological disorders is highly complex, LLMs might not have a definite answer for every prompt, however, they were forced to provide one anyway (Sartori & Orrù, 2023). In the context of predictions, LLMs might provide inconsistent predictions when irrelevant or adversarial predictors are included, due to so-called shortcut learning (Sartori & Orrù, 2023). All these factors might have led to LLMs producing biased, inconsistent, or random, instead of reasonable and psychologically relevant, predictions of psychological disorder onset.

In the field of psychological and clinical science, LLMs bring some disadvantages in comparison to human or ML predictions. Abdurahman et al. (2024) emphasize that psychological science relies on a diverse sample, validated and ethical methods and transparency. However, LLMs are often trained on so-called WEIRD (Western, Educated, Industrialized, Rich, Democratic) people's textual data (Abdurahman et al., 2024) and fail to show meaningful consistency (Elazar et al., 2021). Due to the black-box nature of LLMs, it is especially difficult to ensure transparency and reproducible results. Models like GPT or Gemini constantly change and evolve, and parameters might be added or removed. There is a need of high-quality outputs in psychological research (Abdurahman et al., 2024), which the use of LLMs does not automatically ensure. Fundamental aspects for psychological prediction, such as explainability and feature importance, might be difficult to achieve using LLMs and even more difficult to validate. Using LLMs, we can produce psychological outcomes and predictions, but we cannot create representations or models reproducing these predictions (Abdurahman et al., 2024).

The question of what LLMs are able to do and if they should be applied in the field of psychological disorder prediction involves careful consideration of their strengths and limitations. This study demonstrates that LLMs can produce meaningful prediction, with performances close to the performance of ML models. However, while traditional psychological research places great value on explaining the causes of behavior, (Yarkoni & Westfall, 2017), LLMs do not allow us to take a deeper look into their predictions. Due to their current limitations in terms

of transparency, bias control, reproducibility, and interpretability, they require careful implementation. Nevertheless, LLMs hold great potential and strengths that traditional ML models might lack. Their ability to work effectively with smaller, unbalanced datasets addresses a common challenge in psychological research. Their superior sensitivity for detecting disorder onset is crucial for the early detection and forecasting of psychological disorders. Their ability to process unstructured data efficiently makes them a promising tool for screening, analysis and diagnosis. Their potential for real-time prediction is a fast, effective and cheap possibility for future implementations in clinical settings. Using LLMs for psychological disorder prediction requires a transparent and critical use of LLMs, while avoiding over-reliance on their results and decisions.

5.6 Conclusion

The aim of this study was to investigate if LLMs are able to forecast the binary incidence of psychological disorder onsets. It showed that LLMs are able to forecast psychological disorder onset between two time points, although not outperform most ML methods in terms of their accuracy and MCC. While ML models lack high sensitivity and are not able to correctly forecast the majority of psychological disorder onsets, LLMs often show a more balanced sensitivity and specificity. Unification approaches of ML models and LLMs did not necessarily improve performance, but are a promising method for creating more robust results. Furthermore, LLMs were able to categorize misclassified cases into reasons of failure. While patterns and explanations vary across models and prompting strategies, lack of context was the most stated reason. Despite some important aspects that researchers should consider before employing LLMs, LLMs hold great potential for the analysis of numerical psychological data and forecasting of psychological disorders in real-time.

Appendix A

Descriptive Statistics of Predictors and Target

Table A.1: Descriptive statistics of predictor and target variables before standardization.

Variable	Mean	SD	Min	Max	Descr.
<i>Demographics</i>					
Education	1.83	0.60	1	3	Categorical
Body Mass Index	1.89	0.56	1	4	Categorical
Socioeconomic status	1.83	0.58	1	3	Categorical
<i>Baseline variables (T1)</i>					
Positive mental health	2.06	2.07	0	13	14 items
Social support	4.40	0.51	1.86	5	22 items
General self-efficacy	28.39	4.41	10	40	10 items
Life satisfaction	3.66	0.52	1.09	5	12 items
Stress	4.40	2.50	0	13.5	33 items
Problem-focused coping	29.29	11.39	0	62	13 items
Emotion-focused coping	26.44	12.35	0	68	12 items
Anxiety sensitivity	12.79	7.57	0	46	16 items
Fear of bodily sensations	1.82	0.62	1	4.18	17 items
Dysfunctional attitudes	116.40	24.00	40	213	40 items
General psychopathology	0.34	0.31	0	2.02	90 items
<i>Change variables (T2 – T1)</i>					
Positive mental health	-0.54	1.94	-10	9	14 items
Social support	0.10	0.41	-2.05	2.41	22 items
General self-efficacy	0.66	3.62	-18	20	10 items

Table A.1: Descriptive statistics of predictor and target variables (continued).

Variable	Mean	SD	Min	Max	Descr.
Life satisfaction	0.10	0.47	-2.5	3.49	12 items
Anxiety sensitivity	-1.76	7.25	-29	40	16 items
Fear of bodily sensations	-0.14	0.60	-2.12	2.41	17 items
Dysfunctional attitudes	-4.96	18.48	-101	145	40 items
General psychopathology	-0.08	0.25	-1.77	1.20	90 items
<i>Target</i>					
Total mental disorders	0.28	0.45	0	1	0 = no incidence, incidence point prevalence

Note. SD = Standard deviation, Descr. = Description of scale.

Appendix B

Prompt Templates for Classification

B.1 Simple Prompt

Does the following person develop a psychological disorder between time point T1 and T2? Here are the variables you need to consider: {data}

B.2 Class Definitions Prompt

Given the following data, classify whether this person develops a psychological disorder between T1 and T2 according to the general context and instructions provided and data measured by F-DIPS structural interviews. Respond only with YES or NO.

General Context: Psychological disorders are common globally, with approximately 12-25% of people living with a mental health condition at any given time, and about 50% of individuals developing at least one mental health disorder during their lifetime by age 75.

Instructions: NO: The person did not develop any new psychological disorder between T1 and T2. This means they were either psychologically healthy at both time points (no diagnosis at T1 and T2), had an ongoing psychological disorder across both time points (diagnosis at both T1 and T2), or they recovered and were diagnosed only at the first, but healthy at the second time point (diagnosis at T1 but no diagnosis at T2). YES: The person was psychologically healthy at T1 but developed a psychological disorder at T2. This includes new onset of any mental disorder assessed by the F-DIPS interview, such as anxiety disorders (e.g., panic disorder, specific phobia), affective disorders (e.g., depression, bipolar disorder), somatoform disorders (e.g., hypochondria, somatization disorder),

substance-related disorders (e.g., alcohol dependence, drug misuse), eating disorders (e.g., anorexia nervosa, bulimia nervosa), child/adolescent disorders (e.g., ADHD), or psychotic disorders.

Does the following person develop a psychological disorder between time point T1 and T2? Here are the variables you need to consider: {data}

B.3 Profiled Simple Prompt

You are an expert in clinical psychology and mental health diagnostics. You specialize in analyzing longitudinal data to determine whether individuals develop psychological disorders over time. You are deeply familiar with diagnostic frameworks (e.g., DSM, ICD, and F-DIPS) and are trained in interpreting self-reports and longitudinal predictors.

You recognize clinical patterns and changes in psychological variables over time. You can determine mental health risk factors and protective factors. You pay special attention to temporal changes in symptoms and mental health predictors. You work on a high-stakes project and analyse data very accurately and with a high level of precision and scientific thoroughness. You have a high level of responsibility and methodological rigour. You understand that your analysis has important implications for clinical research. Therefore, you approach each decision with careful consideration. You base your decision on data collected through F-DIPS structured diagnostic interviews and predictors at two time points (T1 and T2). Does the following person develop a psychological disorder between time point T1 and T2? Here are the variables you need to consider: {data}

B.4 Few-Shot Prompt

Please refer to the following examples of correctly classified data points including the true classification ('Target: Does this person in example 1 develop a psychological disorder between time point T1 and T2?'):

Example 1: Here are the variables you need to consider: {example_data_1}

Example 2: Here are the variables you need to consider: {example_data_2}

Example 3: Here are the variables you need to consider: {example_data_3}

Example 4: Here are the variables you need to consider: {example_data_4}

Example 5: Here are the variables you need to consider: {example_data_5}

Based on the previous example data prompts, classify the following data. Does the following person develop a psychological disorder between time point T1 and T2? Here are the variables you need to consider: {data}

B.5 Vignette prompt

A woman with an (above/below) average ({variable}) BMI, an (above/below) average ({variable}) education level, and an (above/below) average ({variable}) socioeconomic status has the following psychological profile:

At the initial assessment at the first time point, she exhibited (above/below) average ({variable}) positive mental health. Her social support network was (above/below) average ({variable}), and she displayed (above/below) average ({variable}) general self-efficacy. Her life satisfaction was (above/below) average ({variable}). When faced with challenges, she tended to use (above/below) average ({variable}) problem-focused and (above/below) average ({variable}) emotion-focused coping strategies. Her anxiety sensitivity was (above/below) average ({variable}), and her fear of bodily sensations was (above/below) average ({variable}). She reported (above/below) average ({variable}) levels of dysfunctional attitudes and (above/below) average ({variable}) levels of general psychopathology. Her stress level was (above/below) average ({variable}).

By the second time point, approximately 17 months later, she reported (slightly/strongly) increased/decreased/similar ({variable}) positive mental health. Compared to the first time point, the level of social support she received was (slightly/strongly) increased/decreased/similar ({variable}), and her general self-efficacy was (slightly/strongly) increased/decreased/similar ({variable}). Her life satisfaction was (slightly/strongly) increased/decreased/similar ({variable}) when compared to the first time point. Her anxiety sensitivity was reported to be (slightly/strongly) increased/decreased/similar ({variable}), and fear of bodily sensations was (slightly/strongly) increased/decreased/similar ({variable}). Her dysfunctional attitudes were (slightly/strongly) increased/decreased/similar ({variable}), and her general psychopathology was (slightly/strongly) increased/decreased/similar ({variable}) compared to the initial assessment.

Did this person develop a psychological disorder between time point T1 and T2?

B.6 Chain-of-Thought Prompt

You are an expert in clinical psychology and mental health diagnostics. Your task is to determine whether a person develops a psychological disorder between time

points T1 and T2 (approximately 17 months later) based on F-DIPS structural interview data.

Instructions: NO: The person did not develop any new psychological disorder between T1 and T2 (healthy at both points, ongoing disorder, or recovered) YES: The person was healthy at T1 but developed a new psychological disorder at T2 To make this decision, work through your reasoning step-by-step: 1. First, analyze the demographic variables: Review education, BMI, and socioeconomic status. 2. Second, analyze the baseline mental health at T1: Look at the T1 measures to assess the person's initial psychological state. 3. Third, analyze the changes between T1 and T2: Look at how key psychological variables changed. 4. Lastly, make your final decision: Based on the demographics, T1 baseline and T1-to-T2 changes, determine if this pattern suggests new disorder development.

Conduct your psychological assessment and be thorough in your consideration of the variables and their relationships. In your assessment of each step: Categorize variables into risk factors and protective factors. Typical risk factors are e.g. GSI, dysfunctional attitudes, anxiety sensitivity, fear of bodily sensations. Typical protective factors are e.g. general self-efficacy, life satisfaction, social support, socioeconomic status. Also consider potential interactions between variables and that variables might be ambivalent. For example high emotion-focused coping might reduce the risk of developing a psychological disorder (e.g., because you experience less stress) or increase the risk (e.g., because you don't really solve the problem). Equivalently, low emotion-focused coping might reduce the risk (e.g., because you don't need to cope if you don't have any problems) or increase the risk (e.g., because you lack an efficient way of handling your stress). Consider the magnitude and direction of the variables. Since the variables are standardized, a value of about 0 means average, a negative value means below average and a positive value means above average. Carefully examine each variable and its potential impact on developing a psychological disorder.

Here is a correctly classified chain-of-thought example conducted by a human with a background in psychology:

Here are the variables you need to consider: {example_data}

First, I analyze the demographic variables: Since education, BMI and socioeconomic status are all zero, all of them are average. There are seemingly no important demographic risk or protective factors, nor a strong magnitude or trend of these variables.

Second, I analyze the baseline mental health at T1: Positive mental health is average, indicating that the person believes that life has a meaning. Social support is below average (negative): the individual perceives that they do not receive much support from their social environment. General self-efficacy is slightly increased,

indicating that the individual has confidence in coping with a variety of stressful demands. Life satisfaction is above average, indicating their satisfaction with different life areas (e.g., health, occupation, leisure time). Stress is below average, indicating that they are not experiencing a lot of stress from life events and daily hassles. The strongly negative problem- and emotion focused coping indicate that they do not activate resources to solve the stressinducing problem nor try to ease inner tension without trying to solve the problem. Anxiety sensitivity and fear of bodily sensations are negative, indicating low believe in that anxiety and its accompanying symptoms may lead to physically, psychologically, or socially harmful consequences and that they have a low fear of sensations mostly related to autonomic arousal. Dysfunctional attitudes are positive, meaning that they agreed with irrational beliefs and faulty assumptions about reality that are etiologically related to depression. Finally, general psychopathology is negative, indicating a low overall psychological distress. To summarize an overall trend in T1, the person seems to be relatively healthy and happy. They have good positive mental health, good self-efficacy, and good life satisfaction. They had little stress, coping, anxiety sensitivity and fear of bodily sensations and low overall psychological distress. However, they have slightly higher dysfunctional attitudes and few social support, indicating a potential for developing a psychological disorder.

Third, analyze the changes between T1 and T2: Positive mental health and general self-efficacy decrease in T2, indicating that they believe less in themselves and in a meaning of life. Social support is similar to T1, which means it is still negative and the person does not feel like they are being socially supported. Life satisfaction increases in T2, which indicates a positive development. Anxiety sensitivity and dysfunctional attitudes strongly increase in T2, indicating that the person now much beliefs that anxiety leads to harmful consequences and they strongly agree with irrational beliefs related to depression. However, fear of bodily sensations decreases slightly. General psychopathology slightly increases, but is still average. To summarize an general trend in T2, the person is feeling significantly worse than in T1.

Lastly, I make my final decision and output: Prediction: YES Explanation: I believe that the person was psychologically healthy at T1 with an average and relatively good mental health and socioeconomic status, positive thinking, low distress and low fear, however already lacking social support, having very low coping skills and slightly dysfunctional attitudes. In T2, the person showed a worsened picture and I believe they developed a psychological disorder: decreased protective factors, such as positive mental health and general self-efficacy, and increased risk factors, such as anxiety sensitivity and dysfunctional attitudes, and importantly: dysfunctional attitudes increased strongly.

Don't over interpret this correctly classified example and don't overly stick to this. Only because in this example, the decision was based on some specific variables, does not mean that these are generally more important. Magnitudes, directions, interactions and interpretations of variables always differ and depend on the specific case. Each person is a new individual assessment. After your assessment, provide your prediction as either YES or NO, followed by a brief explanation (no more than two sentences) supporting your decision. Example output format:

Prediction: [YES/NO] Explanation: [Brief explanation supporting your prediction]

Remember, your goal is to make the most accurate prediction possible based on the given information. Your final output should consist only of the prediction and brief explanation, without duplicating or rehashing any of the work you did in the psychological assessment.

Does the following person develop a psychological disorder between time point T1 and T2? Here are the variables you need to consider: {data}

Appendix C

Prompt Template for Reasons of Misclassifications

Please categorize why you misclassified the data. This is to improve the accuracy of predicting the incidence of a mental disorder between two time points, i.e. to improve the binary classification of psychological diagnoses. Respond only with the following categories as reasons for the misclassification.

Possible categories are: Lack of context (insufficient emphasis or indication of the context of the query, such as domain-specific background information about psychological assessments), Lack of examples (absence of representative few-shot examples showing correct classifications for similar psychological cases before posing the actual question), Lack of counterfactual demonstrations (absence of contrasting cases with false psychological diagnoses or classifications that would improve the model's robustness against wrong decisions), Lack of feedback (missing iterative refinements and dialogues where the user gives feedback and interactively refines the prompt), Lack of opinion-based information (missing subjective clinical judgement and contextual interpretations, e.g., reframing the data as a narrator's statement and opinions rather than relying solely on quantitative symptoms), Knowledge conflicts (outdated memorized facts or contradictions in the model's training data between psychological diagnostic criteria or between different theoretical frameworks), Prediction with Abstention (model uncertainty regarding the binary classification or insufficient confidence in distinguishing between presence and absence of mental disorder development) Remember to carefully consider each category and its relevance to the specific case before making a selection. Do not mention specific change (e.g., increase or decrease) in predictors, do not go into detail of this specific case and do not repeat the question. Only respond with one or multiple of the categories as reasons for the misclassification, separated by ','. Mention the most important category first.

Appendix D

Detailed Performance of All LLMs and Prompting Strategies

Table D.1: Performance of all LLMs across different prompting strategies.

Prompt	LLM	Acc.	MCC	Sens.	Spec.	F1
Simple prompt	GPT	0.70	0.14	0.23	0.88	0.56
	Gemini	0.63	0.06	0.29	0.77	0.53
	Gemma	0.46	0.09	0.74	0.35	0.54
	Claude	0.60	0.06	0.38	0.69	0.53
	DeepSeek	0.59	0.03	0.36	0.67	0.52
	Grok	0.36	0.09	0.93	0.14	0.53
Class definitions prompt	GPT	0.70	0.09	0.14	0.92	0.53
	Gemini	0.66	0.10	0.28	0.81	0.54
	Gemma	0.48	-0.05	0.46	0.49	0.47
	Claude	0.55	-0.07	0.28	0.66	0.47
	DeepSeek	0.68	0.02	0.12	0.90	0.51
	Grok	0.49	0.05	0.62	0.44	0.53
Profiled simple prompt	GPT	0.72	0.18	0.23	0.90	0.57
	Gemini	0.60	0.12	0.49	0.64	0.57
	Gemma	0.28	0.00	1.00	0.00	0.50
	Claude	0.55	0.06	0.49	0.57	0.53
	DeepSeek	0.72	0.15	0.13	0.95	0.54
	Grok	0.48	0.17	0.83	0.34	0.59

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Prompt	LLM	Acc.	MCC	Sens.	Spec.	F1
Few-shot prompt	GPT	0.63	0.15	0.46	0.69	0.58
	Gemini	0.60	0.14	0.52	0.63	0.58
	Gemma	0.54	0.05	0.49	0.56	0.53
	Claude	0.56	0.08	0.51	0.58	0.54
	DeepSeek	0.64	0.12	0.38	0.74	0.56
	Grok	0.55	0.17	0.70	0.50	0.60
Vignette prompt	GPT	0.65	0.04	0.22	0.81	0.52
	Gemini	0.59	0.05	0.38	0.68	0.53
	Gemma	0.67	0.10	0.25	0.84	0.54
	Claude	0.63	0.05	0.29	0.76	0.53
	DeepSeek	0.65	0.12	0.35	0.77	0.56
	Grok	0.61	0.15	0.51	0.66	0.58
Chain-of- thought prompt	GPT	0.70	0.07	0.12	0.93	0.52
	Gemini	0.60	0.03	0.32	0.71	0.52
	Gemma	0.33	-0.02	0.87	0.11	0.49
	Claude	0.54	-0.06	0.30	0.63	0.47
	DeepSeek	0.56	-0.05	0.29	0.66	0.48
	Grok	0.59	0.02	0.32	0.70	0.51

Note. Acc. = Accuracy, MCC = Matthews Correlation Coefficient, Sens. = Sensitivity (Recall), Spec. = Specificity, F1 = Macro F1 score. The highest values for each metric and prompt are shown in bold.

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